



Attitudinal Tensions in the Joint Pursuit of Explainable and Trusted AI

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Abstract

It is frequently demanded that AI-based Decision Support Tools (AI-DSTs) ought to be both explainable to, and trusted by, those who use them. The joint pursuit of these two principles is ordinarily believed to be uncontroversial. In fact, a common view is that AI systems should be made explainable *so that* they can be trusted, and in turn, accepted by decision-makers. However, the moral scope of these two principles extends far beyond this particular instrumental connection. This paper argues that if we were to account for the rich and diverse moral reasons that ground the call for explainable AI, and fully consider what it means to “trust” AI in a descriptively rich sense of the term, we would uncover a deep and persistent tension between the two principles. For explainable AI to usefully serve the pursuit of normatively desirable goals, decision-makers must carefully monitor and critically reflect on the content of an AI-DST’s explanation. This entails a deliberative attitude. Conversely, calls for trust in AI-DSTs imply the disposition to put questions about their reliability out of mind. This entails an unquestioning attitude. As such, the joint pursuit of explainable and trusted AI calls on decision-makers to simultaneously adopt incompatible attitudes towards their AI-DST, which leads to an intractable implementation gap. We analyze this gap and explore its broader implications: suggesting that we may need alternate theoretical conceptualizations of what explainability and trust entail, and/or alternate decision-making arrangements that separate the requirements for trust and deliberation to different parties.

Keywords AI-based Decision Support Tools · Decision-making · Explainability · Trust

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1 Introduction

Contemporary discussions about ethical and socially-beneficial AI often refer to two key principles: AI systems should (a) be made *explainable* to humans, and (b) be *trusted* by their users. Although recent scholarship has paid much attention to practical questions of how to make AI systems explainable and how to build trust in them, few have stopped to ask: can these two principles be meaningfully pursued together?

To some, this question is a non-starter: these two principles not only can, but *must* be jointly pursued, for explainability is a key factor for ensuring trust in AI. Indeed, a commonly held view is that AI systems should be made explainable *so that* we can build trust in these systems, and in turn, so that we can ensure their widespread adoption and acceptance (Chatila et al., 2021; Durán & Formanek, 2018; Deloitte, 2021; European Commission 2019; ISO/IEC, 2020; Stanton & Jensen, 2021). Hereafter, we will call this view the ‘process model’.¹ Insofar as we only care about the explainability of AI to the extent that it leads to trust, the process model seems unproblematic. However, as we will discuss, the moral scope of these two principles extends far beyond this particular instrumental connection. Specifically, we care about explainability because an AI system’s explanations might help us pursue a variety of morally desirable goals beyond trust: fairness, accountability, and legitimacy, to name a few. Here is the problem. Even if the process model is correct, it remains to be asked: what happens *after* we obtain trust and, finally, acceptance? Can explanations still meaningfully serve their expected normative purpose even after trust is established?

In this paper, we suggest that if we were to account for the rich and diverse moral reasons that ground the call for explainable AI, and consider what it means to “trust” AI in a descriptively rich sense of the term, we would uncover a deep and persistent tension between the two principles. The problem, as we shall explain, is that these two principles give rise to incompatible demands about the appropriate attitude that users should adopt towards their AI systems. For the explanations of AI systems to be useful for the various normative ends they are expected to serve, users must adopt a deliberative attitude. However, trust in AI systems entails that users adopt an unquestioning attitude – one that is incompatible with the deliberative attitude.

Uncovering the exact nature of these attitudinal tensions between explainability and trust, however, requires considerable theoretical groundwork. We lay some of this groundwork in Sect. 2, where we explore in greater detail what the popular calls for explainability and trust in AI entail. Section 3 elaborates on the process model, and briefly highlights its shortcomings. Section 4 follows by making explicit the attitudinal tensions between explainability and trust. Finally, in Sect. 5, we trace the implications of our account, and explore a few theoretical and practical workarounds.

¹ We use this particular term in order to make explicit the connection to similar ‘process models’ developed in the fields of information systems and organizational behavior (e.g., the ‘Technology Acceptance Model’) – where the pursuits of trust, ease of use, result demonstrability, etc. are justified exclusively in terms of their instrumental utility in obtaining the clearly defined end-goal of ‘acceptance’. We discuss these process models and further elaborate on our choice of this term in Sect. 3.

2 AI Systems in Decision-Making Contexts

As a general technological category, artificial intelligence – “AI” – has been instantiated across a variety of domains including medicine, business, and government; and in configurations that encompass robotic aids, virtual chatbots, and search and recommendation systems, to name but a few. In this paper, we focus on one type of AI system, specifically, AI-based Decision Support Tools – hereafter “AI-DSTs” – that output recommendations intended to support human decision-making, often as part of an integrated human-AI workflow.

AI-DSTs are, in our view, emblematic examples of close human-AI collaboration. Their implementation can be found in clinical settings for optimizing diagnosis and pharmaceutical treatment plans, in organizations to streamline hiring decisions by managers, and in government to help decision-makers craft policy using large amounts of data, to name but a few. Typically, AI-DSTs’ recommendations are received by users – sometimes called “humans-in-the-loop” (Zanzotto, 2019) – who then accept or reject the proposed action as a contribution to the decision-making workflow. The success of these final decisions therefore turns on the ability of this decision-maker to determine the suitability of the AI-DST’s recommendation against a combination of their own expertise, experience, and judgement.

In return, decision-makers expect AI-DSTs to make reasonable inferences from large amounts of data in such a way that complements their workflow. Unreasonable expectations of an AI-DST’s capabilities and limits can lead to these tools adding to the decision-makers’ cognitive burdens, instead of lessening them. To ensure that these human-AI interactions are successful in improving decision-making workflows, scholars and practitioners have surfaced various proposals for overcoming key problems that could hinder such interactions. The calls for *explainability* and *trust* are two especially prominent examples of such proposals (Floridi et al., 2018; Hagedorff, 2020).

First, the call to make AI-DSTs explainable may be seen as responding to worries about the “opaqueness” of these systems that might prevent decision-makers from understanding how their AI-DSTs work, and in turn, from effectively collaborating with them. AI-DSTs can be opaque – that is, the internal logic by which they process inputs to generate outputs can be unknown to decision-makers – for various reasons. A few such reasons are common to most decision-support technologies: sometimes the decision-maker might not have the requisite technical literacy to understand how the system works, or sometimes those who developed and deployed the system might intentionally keep its inner logic secret (Pasquale, 2015). Nevertheless, a certain kind of opacity seems unique to AI systems, one that arises out of a “mismatch between mathematical optimization in high-dimensionality characteristic of machine learning, and the demands of human-scale reasoning and styles of semantic interpretation” (Burrell, 2016, p. 2).² Some AI-DSTs are “black-boxes” precisely because they are

² Although we use terms like “AI system” and “AI-DST” throughout our argument, it is important to note that worries about opacity are most salient when AI models are trained using certain machine learning (ML) techniques, since ML models tend to be architecturally more complex, and hence less intelligible to humans (Gunning et al., 2017). However, partly because even simple regression models can be opaque in certain interesting ways (see, for example, Lipton’s (2016) discussion of opacity arising from feature

designed to locate complex patterns in large and highly-dimensional datasets that no human could reasonably be expected to parse (Long, 2020).

Second, the call for trust in AI may be seen as responding to worries that decision-makers could prematurely discount their AI-DST's recommendations, and in turn, fail to fully integrate these systems into their decision-making processes. Organizations typically need to expend considerable resources and effort to implement and integrate AI-DSTs – or indeed, any decision-support tool – into decision-making workflows. Consequently, organizations are strongly motivated to ensure that decision-makers fully utilize their AI-DSTs, to offload some of their cognitive burdens and decision-making tasks so that they might make better, if not more efficient decisions. However, if decision-makers do not trust their AI-DST, such benefits are unlikely to be realized. In line with these pragmatic considerations, conversations about trust in AI-DSTs often center on considerations such as ease of use, system performance and reliability, and other such factors that might affect the quality of human-AI collaboration.

2.1 Popular Calls for Explainable and Trusted AI

These two proposals have, over time, come to be connected to larger moral and practical questions surfaced in scholarship on ethical and socially-beneficial implementations of AI-DSTs. Grappling with problems of explainability and trust not only furthers the effectiveness of human-AI collaboration, but also ties into larger questions of how AI-supported decision-making can be made more fair, accountable, robust, and legitimate. As such, the rest of this section expands on the importance of explainability and trust as desirable principles to pursue in developing and deploying AI-DSTs.

2.1.1 Unpacking the Calls for Explainable AI-DSTs

Calls for explainability ask that the inner logic by which AI-DSTs process inputs to generate outputs should be made available and understandable to human decision-makers. Computer scientists and engineers have responded enthusiastically to these calls. In recent years, AI and human-computer interaction research has seen considerable progress in the development of various explainability techniques to make the inner logic of AI systems accessible to human users. Various such explainability techniques can be used to present – through text, visualizations, graphs, networks, and other diagrams – information about an AI-DST's features, weights, probabilities, and other relevant parameters, through which a decision-maker can come to at least a basic understanding of the system's inner logic (for a recent review of XAI techniques, cf. Mueller et al., 2019). Other approaches – such as the provision of “counterfactual” or “contrastive” explanations (Nguyen et al., 2015; Wachter et al., 2017) – aim to provide such knowledge without opening the “black-box”, by providing counterexamples of how the system would have behaved under different scenarios.

selection and preprocessing), and partly because ML techniques are increasingly commonplace in modern AI deployments, we decided that it would be appropriate to use the more general terms (AI/AI-DST) throughout.

Of course, there remains debate over whether we can truly achieve perfect knowledge about what exactly is going on inside an AI system (Humphreys, 2004; Lipton, 2019), whether explainability necessarily comes at the cost of accuracy (Gunning, 2017; Rudin, 2019), and whether it is possible to meaningfully intervene to prevent potentially harmful decisions simply by looking at the system's explanations (Cappelen & Dever, 2021; Robbins, 2019). Moreover, despite the availability of several explainability techniques, organizations have been slow to implement these techniques in their AI-DST deployments (Benbya et al., 2020; Deloitte, 2021). These debates and problems notwithstanding, with growing research on the explainability and interpretability of AI, decision-makers today have available to them several technical possibilities of understanding, to some extent, how their AI-DST came to its recommendations.

This brings us to a question of key importance for our paper: what exactly do we gain, morally speaking, by making AI-DSTs explainable? One overwhelmingly popular answer to this question is the following: *explanations provide an epistemic warrant for trust* (Pieters, 2011; von Eschenbach, 2021; Ferrario et al., 2022).³ Knowledge about the inner workings of their AI-DST, it is argued, can help decision-makers come to reasoned and well-calibrated judgments about when and why to trust its recommendations. Rather than making their own inferences and predictions about how their system might behave, decision-makers can use explanations to identify the patterns that distinguish AI-DST's correct and incorrect outputs, and in turn, come to better-formed judgments about how to integrate these tools into their decision-making (Simon, 2010; Jacovi et al., 2021). Along these lines, explanations might be framed as reasonably acceptable if their fidelity and clarity provide sufficient evidence supporting a *past* recommendation, so that decision-makers come to trust *future* recommendations.

There remain lively debates about precisely how much, and what kinds of information needs to be communicated about an AI-DST's inner logic to instill in a decision-maker the right kind of trust: i.e., trust in future outputs (Durán & Formanek, 2018). However, few question the straightforward manner by which explainability and trust in AI-DSTs seem to be connected to one another. Ostensibly, explanations can lead to trust, and in turn, trust can be treated as a construct that measures the effectiveness of rendered explanations.

³ Ferrario et al. (2022) provide an especially detailed and careful account of how explainability contributes to trust in AI. Specifically, they argue that explainability fosters trust if and only if it (a) provides justification for a belief about the trustworthiness of the AI system, and (b) causally contributes to rely on the AI in the absence of monitoring. Interestingly, in developing their arguments, they employ a concept of "paradigmatic trust", one component of which is an "anti-monitoring" stance towards the trustee. As we will subsequently discuss, this view relates very closely how trust is conceptualized in the present paper (i.e., as comprising an 'unquestioning attitude'). For this reason, one may reasonably see our paper as a natural extension to Ferrario et al.'s arguments – while they focus on the attitudes, beliefs and dispositions that make explanations useful for fostering trust in AI, we focus on what happens (at the attitudinal level) *after* this trust is fostered. Specifically, they ask: *when does (and doesn't) explainability foster trust?* And, we ask: *can explanations still serve their intended normative purpose after trust is fostered?* Our thanks to an anonymous reviewer for pointing out this interesting connection.

However, there are other popular, and deeply important answers to the question of how explanations from AI-DSTs improve decision-making.⁴ For one, explanations can help decision-makers perform regular “sanity checks” to ensure that their AI-DST is functioning as intended, and that its internal logic is generalizable and robust (Ribeiro et al., 2016). In case the AI system performs inconsistently and/or inaccurately, scrutinizing explanations can help the decision-maker carefully evaluate the overall capability of their AI-DST to render appropriate recommendations. Moreover, explanations might also help decision-makers identify biases in algorithmic decisions: to detect and intervene in systems that unfairly harm (or privilege) certain groups over others (Sonboli et al., 2021). Yet another popular and important proposal suggests that explanations are important for decision-makers to preserve “meaningful control” over AI systems, so that the responsibility for AI-assisted decisions can be retained in human hands (Coeckelbergh, 2020). There are also important *epistemic* reasons for making AI-DSTs explainable. By analyzing how complex AI models make decisions, decision-makers may be exposed to new – and perhaps more efficient and reasonable – decision-making strategies, which they might subsequently incorporate into their own decision processes (cf. Robbins, 2019). For such reasons, despite some open questions about why exactly explainable AI matters morally, it remains relatively uncontested that, when we insist that AI-DSTs should be made explainable, decision-makers are better placed to pursue decisions that accord with fairness, meaningful human control, robustness, efficiency, and various other normatively desirable goals.

2.1.2 Unpacking the Calls for Trust in AI

Popular calls for trust in AI have often taken care to acknowledge that only certain AI systems deserve to be trusted. The fact that one is far more likely to encounter calls for ‘trustworthy AI’ than ‘trusted AI’ is a testament to this carefulness: scholars and policymakers alike have been warning against encouraging trust in AI systems that are biased, harmful, or otherwise undeserving of trust (Floridi, 2019; Chatila et al., 2021), and/or that decision-makers might place too much weight on AI-DST’s recommendations, even when they ought not to do so (cf. Cummings, 2017). Yet, at least in some cases – when AI-DSTs are carefully designed to be lawful, ethical, and robust (European Commission, 2019), when decision-makers are unduly biased against accepting their AI-DST’s recommendations (Dietvorst et al., 2015), or when the decision-making task at hand requires certain data-analysis and prediction tasks that an AI system could perform much more efficiently than a human decision-maker (cf. Killock, 2020; Sreshta et al., 2019) – it seems appropriate to call on decision-makers to take their AI-DST’s recommendations more seriously. In such cases, calls for *trust* in AI seem reasonable.

⁴ Explanations from AI, of course, are not only useful for decision-makers to improve their decision-making. There are a number of important moral reasons for pursuing explainability that center on the value of explanations to *decision-subjects* – to contest and seek recourse for unfair decisions, to give informed consent to AI-driven decision-making processes, etc. We will return to discuss these in Sect. 5 [Implications].

However, it is difficult to pin down what exactly this ‘trust’ entails. This is in part because the concept itself is continuously unsettled and contested: trust has been theorized in a variety of intellectual traditions, each with their own unique understanding of what the concept entails. It falls outside the scope of this paper to engage in any substantive debates about the nature of trust. However, to provide some scaffolding for our subsequent analysis, it is useful here to briefly describe important distinctions between different approaches to theorizing trust.

Some views of trust are *morally-loaded*, in that they aim to characterize how a trustor ought to evaluate the trustworthiness of potential trustees, and/or how trustees ought to respond to the trust that is placed in them. On some such views, the typical trustor should trust someone when they observe that this potential trustee possesses certain desirable properties that make them *trustworthy* – benevolence, integrity, reliability, honor, etc. (Hoff & Bashir, 2015; Mayer et al., 1995). Other views suggest that a trustee should take the fact that they are being trusted to do X as a reason to do X, and correspondingly, their trustor can take this expectation of goodwill as a reason to trust them (Jones, 1996). Importantly, most moral views of trust are *agent-directed*. That is, moral trust usually involves the ascription of certain agential states and/or properties to both trustors and trustees, that only human agents can be said to reasonably possess (Nickel, 2010). This makes such views relatively unhelpful for characterizing trust in AI, since non-agential technological artifacts cannot reasonably be described as having intentions, consistently adhering to principles, or being motivated by goodwill towards trustors. As such, if artifacts cannot possess any of the properties that trust, in a morally-loaded sense of the term, is based on, then ‘trust’ in an AI system is either mistaken (in that it ascribes to the system agential properties that it does not possess), or misdirected (in that it confuses trust in the system’s human developers and deployers for trust in the system itself). As Ryan (2020) persuasively puts it, “AI should not be viewed as trustworthy because [this] undermines the value of interpersonal trust, anthropomorphizes AI [...] and diverts responsibility from those developing and using AI” (p. 2750).⁵

However, it is also possible to conceptualize trust at a *descriptive* level. Here, the emphasis is on accurately and completely characterizing the attitudes and behaviors that the typical trustor adopts towards a potential trustee. As such, descriptive characterizations of human-AI trust would focus on describing the attitudes that a human decision-maker might adopt towards an AI-DST that they trust, and the factors that influence these trust-based interactions. There is a rich body of empirical literature – especially in the fields of psychology and human-computer interaction – that attempt to answer such questions (cf. Lee & See, 2004). Here, trust has been found to variously depend on dispositional factors (such as the personality or age)

⁵ Although we happen to find Ryan’s view generally persuasive, the arguments in this paper do not require a firm stance on any particular position about whether AI systems can be trustworthy and/or trusted in the paradigmatic interpersonal senses of these terms. If the reader is persuaded by Ryan’s (and/or other similar) arguments about trust in AI being a form of category error, they might still accept without contradiction that (a) people can adopt similar attitudes and dispositions towards an AI system as they would towards a human they trust, and (b) sometimes this ‘trust’ (in a descriptively rich, but non-normative sense of the term) is desirable. Conversely, if the reader believes that full-blooded normative trust in AI is possible, our arguments would be readily compatible with their view.

of decision-makers; situational factors such as system complexity, task difficulty, and the general environment under which the human-technology interaction takes place; and factors related to how past interactional experiences shape perceptions of future performance, and hence trust (cf. Hoff & Bashir, 2015). Another recent review identifies the AI-DST's "tangibility" (i.e. the physical characteristics of its interface), "transparency" (i.e. access to knowledge about its inner workings), "reliability" (i.e. its record of past performance) and "immediacy behaviors" (i.e. its responsiveness and adaptiveness) as key determinants of trust (Glikson & Woolley, 2020). Such empirical findings certainly help us better understand how decision-makers relate to their AI-DSTs, and what factors might help them make more accurate decisions about when to accept or reject the system's recommendations.

However, it is important to note that most empirical research on this topic conceptualizes 'trust' as a matter of adjusting one's reliance on a system such that one tends to accept correct recommendations and reject incorrect recommendations (Ehsan et al., 2021; Hoff & Bashir, 2015). On this widely-received view, trust in technology is a matter of "appropriate reliance" (Lee & See, 2004) where "the basis of trust [is described] in terms of the performance of an agent, the trustee, who furthers the goals of an individual, the trustor" (p. 54). In our opinion, however, such purely goal-oriented views do not always accord with reasonable intuitions about what trust in AI-DSTs in fact looks like. If, for instance, a decision-maker constantly monitors their AI-DST and expends considerable cognitive effort to painstakingly scrutinize its output and explanations for each decision, we would be inclined to think that they do not trust their AI-DST – even if they end up accepting most of its correct recommendations and rejecting most of its incorrect ones. If this inclination is correct, we ought to prefer a view of trust that describes a trustor's willingness to rely on a trustee without (or with relatively little) monitoring, suspicion, and scrutiny (Taddeo, 2017).

To clarify this discrepancy, it is useful to tap into a widely-accepted philosophical distinction between 'trust' and 'mere reliance'. Baier (1986), who first popularized this distinction, offers the following heuristic to help us separate the two concepts. When we *rely* on someone to do X, and they fail to do X, we feel at most disappointed. In this case, we have no serious grounds to be personally, sharply affected by this failure. However, when we *trust* someone to do X, and they fail to do X, we feel *betrayed*. Trust, being more normatively loaded than reliance, contains the possibility of normative failure, and hence betrayal. As such, to fully describe the attitudinal orientation that some trustor (say, some decision-maker) adopts towards a trustee (their AI-DST), we must be able to account for the possibility of betrayal – and its accompanying sharp emotional response – in cases of failure.⁶

⁶ Baier ultimately defends a morally-loaded view of trust: she believes that a trustor's assumption of *good-will* on part of their trustee is what generates betrayal during failures, and in turn, is what separates trust from mere reliance. As mentioned before, we wish to keep the arguments in this paper neutral between specific conceptualizations of trust – since our focus is on characterizing the *attitudes* that the typical trustor adopts towards a trustee. As such, we are taking Baier's distinction between trust (betrayal) and reliance (disappointment) as an attitudinal marker to separate the two concepts in terms of the differing attitudes they evoke in trustors (or reliers). This usage of Baier's distinction is not especially idiosyncratic – Holton (1994) is a notable example of an attempt to theorize trust using Baier's trust-reliance distinction as an attitudinal marker.

Is it possible to conceptualize trust in AI at a descriptive level while retaining a sensitivity to the trust-reliance distinction? We believe so. To generate this intuition, consider the following scenario. Suppose a laptop, one that has been in our possession for several years and functioned perfectly throughout this period, suddenly fails in the middle of drafting a manuscript. At this critical moment of failure, it seems like an understatement to say that we would feel merely disappointed. Rather, we might reasonably experience a sharp emotional response towards the laptop – perhaps anger of a sort that gestures towards betrayal. Our “trust” – if this is indeed the accurate term – does not seem to be tied to any beliefs about the laptop’s agential properties (to its “benevolence” or “integrity”): it does not seem like we are necessarily anthropomorphizing the laptop. Our trust, further, need not necessarily be tied to any normative expectations from the developers, manufacturers, and various other human agents that were involved in building the laptop and bringing it in contact with us. Rather, our feelings of ‘betrayal’ might be more reliably traced to the long-standing positive cooperative relationship we built with this laptop over years, and our well-founded (but ultimately frustrated) expectation that it would reliably support our manuscript-drafting efforts. We feel betrayed, to borrow Margalit’s (2017) language, when our thick relations with the laptop come unglued (p. 47), and we no longer feel confident depending on it for future writing tasks.

It is not uncommon to encounter such descriptions of thick (intimate, affective) relations between humans and technical artifacts. Empirical studies in human-robot interaction, for instance, have documented the ways in which we form deep attachments with robots, as, over sustained positive interactions, we gradually come to see them as reasonable targets for intimacy, affection, and strong normative expectations (see e.g. Sung et al., 2007; Darling et al., 2015; Mollen et al., 2022). Studies have also documented people ascribing intentionality and other such agential attributes to technical artifacts that they interact with (see e.g., Wiese et al., 2017; Bigman et al., 2019).⁷ Betrayal – in the descriptively rich sense of the emotion – has also been studied in human- technology interactions, especially in cases where technologies *deceive* us about what they can and cannot do (see e.g., Danaher et al., 2020; Saeetra, 2021). In such cases, as Coecklebergh (2012) argues, it does not matter so much whether robots (or other seemingly ‘intelligent’ technical artifacts, like AI-DSTs) are legitimate moral agents – what matters is that they appear to us as more than objects. It is therefore not unreasonable to imagine that a decision-maker might adopt attitudes and behaviors towards an AI-DST they trust that very closely resemble the paradigmatic case of human-human trust. Importantly this ‘trust’ – in a descriptively rich, but non-normative sense of the term – would entail the adoption of quite differ-

⁷ This is especially noteworthy, since, in Simon’s (2010) view, the “ascription of intentionality is crucial for the feeling of betrayal” (p. 347). If some artifact fails at some task it was ‘trusted’ to perform, and one ascribes intentionality in this failure (i.e., that the artifact ‘chose’ to fail its task), they might reasonably feel betrayed. As Simon further argues, “whether [one] feels betrayed or disappointed resides in [one’s] perception of the reasons for failure” (p. 347). These perceptions may be mistaken, and perhaps one ought not to ascribe intentionality to non-agential artifacts: but at a descriptive level, the attitudes and affections of ‘trust’ and ‘betrayal’ look the same.

ent attitudes and dispositions towards the AI-DST than if they were to merely rely on it.⁸

In Sect. 4.2., we will return to develop a more complete view of what such ‘trust’ between a decision-maker and their AI-DST might look like, and why this trust is at least sometimes desirable. For the purposes of this section, however, it should suffice to appreciate that trust and reliance entail quite different attitudinal states, and, whether or not we can ‘trust’ AI in the same morally-laden way as we would another person, it is at least plausible to think that we could reasonably adopt trust-like attitudes and behaviors towards AI systems in ways that go beyond mere reliance.

3 A Process Model Between Explainability and Trust

Beyond AI-DSTs and AI systems more generally, scholars – especially in the fields of information systems and organizational studies – have proposed various models that aim to explain and predict the effective integration of new workplace technologies. Many of these proposals build upon the prominent “Technology Acceptance Model” – or “TAM” – first advanced by Davis (1989), in which he proposes and validates two theoretical constructs, *perceived usefulness* and *perceived ease of use*, to explain both current and predicted use and acceptance of information systems. According to TAM, perceived usefulness is an important factor since “people tend to use or not use an application to the extent they believe it will help them perform their job better” (Davis, 1989, p. 320). In turn, even if a new technology might be seen as beneficial to future job performance, understanding its perceived ease of use is important since technology rejection may arise from users who “believe that the [system] is too hard to use and that the performance benefits of usage are outweighed by the effort of using the application” (p. 320). Subsequently, Venkatesh and Davis’ (2000) extended model, TAM2, incorporates additional theoretical constructs to include “social influence processes (subjective norm, voluntariness, and image) and cognitive instrumental processes (job relevance, output quality, result demonstrability, and perceived ease of use)” (Venkatesh & Davis, 2000, p. 187). Substantial evidence has been presented in broad support of TAM, and its influence as a *process model* that aims to identify and measure factors in pursuit of a clearly defined end-goal – i.e., technology acceptance – has continued to pervade industry practice and academic scholarship since.

Current industry language around AI-DSTs that describes the basic instrumentality of explainability building trust is strikingly similar to descriptions of TAM and related process models. The need for explainability, in this case, is described as helping users calibrate their trust: to be able to know *when* to “double-check” the recommendations of AI-DSTs, to speed up user onboarding and system familiarity/confidence, and to manage expectations around building user understanding of inscrutable systems. In turn, these calibrations help to further manage the overall influence that AI-DSTs might have on users (Chatila et al., 2021; Floridi, 2019;

⁸ For an instructive discussion of this distinction between descriptive and normative views of trust in non-agential artifacts, and how they are both distinct from reliance, but not equivalent to one another, see Tallant’s (2019) provocatively-titled paper: “You can trust that ladder, but you shouldn’t”.

Google, n.d.; ISO/IEC, 2020; Stanton & Jensen, 2021). The rest of this section examines in greater detail the process model that instrumentally connects explainability to trust, and in turn, to adoption and acceptance.

3.1 Explanations Building Trust; Trust Leading into Adoption

There has been considerable recent research to better understand how explanations build – and sometimes fail to build – trust. Explanations that do not represent the system's inner logic with enough detail can be unduly disregarded (Papenmeier et al., 2019; Shin, 2021; Weitz et al., 2019), but so too can explanations that are overly complex and hence difficult to understand (Bansal et al., 2021; Yang et al., 2017). Decision-makers that have contextual and domain-specific expertise can be more likely to interpret their AI-DSTs explanations accurately, and hence produce correct decisions (Alufaisan et al., 2020; Janssen et al., 2020). Conversely, this would also imply that explanations should be tailored to prioritize certain context- or domain-specific information to complement their users. Yet, the presence of tailored explanations or interpretability tools might also be taken for granted – that is, the superficial *presence* of explanations might stand in for actual scrutiny and deliberation over the substantive content of explanations themselves – which could, in turn, lead to cases of over-trust and misuse (Kaur et al., 2020). These findings, in general, suggest that although explanations can be useful for decision-makers to build trust, the link between the two principles should not be taken to be unproblematic.

Moreover, prior research in the fields of information systems and organizational behavior has also shown that trust in decision-support tools tends to drive positive perceptions about their ease of use and usability (Wu et al., 2011). Similar refrains can be commonly heard in contemporary conversations about trust in AI-DSTs: about how the potential benefits and returns of implementing AI systems will fail to be realized if decision-makers do not trust – and in turn, accept and adopt – their AI-DSTs (Chatila et al., 2021; Gillespie et al., 2020; Stanton & Jensen, 2021). Trust in AI is therefore suggested to be crucial for ensuring that decision-makers fully adopt AI-DSTs into their decision-making processes, and effectively integrate these tools into their decision-making workflows. Explainability, insofar as it leads to trust, is ostensibly desirable for the same reason.

This model of human-AI interaction, which largely follows a similar procedural logic to TAM, seems to be an attractive proposition for thinking through and identifying the factors which might hinder the adoption and acceptance of AI-DSTs. Moreover, the theoretical constructs that such an approach provides are clear ways of operationalizing and measuring human and organizational factors that can be practically addressed and acted upon. In this sense, the utility of the process model should not be discounted. Our contention, however, is that the normative value of these two principles extends far beyond their linear relationship to one another, and in turn, to adoption. On the process model, once explanations serve their intended purpose of building trust, they ostensibly cease to be of value. This view, as we discussed previously, seems rather incomplete: we care about explainability for a variety of important moral goals that do not diminish in importance even after trust is achieved. The inadequacy of the process model to take into account these rich and diverse moral

goals is our primary motivation for moving to shift the meaning of explainability and trust away from their singular definition in terms of technology adoption and acceptance. In so doing, however, we uncover key attitudinal tensions that emerge in the joint pursuit of these two principles.

4 Attitudinal States for Explainable and Trusted AI

Consider the following vignette:

Zoe is using an AI-DST to assist in prescribing medicines to her patients. This system outputs a list of drugs and their prescribed dosing amounts, based on the patient's characteristics and Zoe's diagnosis, among other inputs. Zoe has not been using this system for very long, and does not yet have a good understanding of its strengths and weaknesses. Fortunately, the AI-DST happens to be explainable. She deliberates over each outputted prediction and explanation with great care, and over time, comes to a well-calibrated understanding of when the system tends to get things right and wrong. Over time, she comes to trust that, at least for a certain set of medical diagnoses, the AI-DST is particularly good at prescribing the correct medicines and doses. As a result, when dealing with the diagnoses in this trusted set, she ceases to be suspicious about the AI-DST's reliability and begins to accept its prescriptions without much scrutiny or deliberation. If, after this point, the AI-DST happens to output a gravely erroneous prescription for a diagnosis in the trusted set, Zoe would be unlikely to notice this mistake.

This vignette, while perhaps a little simplistic, illustrates a tension that we will develop for the rest of this section. Zoe's trust in her AI system seems thoughtful and well-calibrated, based on her careful deliberation about its output and explanations. But, when this trust is established, she begins to ignore its explanations, and in turn, misses out on potentially vital information about its biases and errors. This tension, we suggest, is not some accidental feature of our imagined vignette: rather, it inevitably emerges when we pursue normatively full-blooded conceptions of what it means to trust an AI system, and why we ought to make AI explainable.

4.1 Explainability and the Deliberative Attitude

In Sect. 2.1 above, we briefly outlined various morally desirable goals that an AI-DST's explanations can help us pursue: fairness, trust, legitimacy, and accountability, to name a few. Here, we wish to once again reiterate that explanations might in some cases be necessary, but never sufficient for achieving these moral goals. The mere act of making an AI system explainable does not, by itself, guarantee us desirable ethical outcomes. Rather, explanations carry information about an AI system that needs to be scrutinized, reasoned about, and acted on in ways that further moral ends.

Minimally, then, a decision-maker should pay close attention to their AI-DST's explanations: they should carefully monitor their system and proactively engage with the content of its explanations *during* the decision-making process. It would not suffice, for instance, to take for granted the mere fact that their system is explainable, and only retrospectively review its explanations *after* something goes wrong. Rather,

decision-makers should proactively look inside the black-box, so to speak, and incorporate the information contained in their AI-DST's explanations beyond being content with its superficial presence. By way of highlighting the need for such proactive monitoring, consider the popular proposal that explanations are necessary for decision-makers to retain *meaningful control* over (and in turn, be held accountable for) AI-supported decisions. For control to be meaningful, at minimum, decision-makers must have sufficient information about how the decision in question was made, including about the tools that supported the decision-making process (Coeckelbergh, 2020). This epistemic condition for meaningful control would clearly be unmet in cases where the decision-maker failed to monitor their AI-DST and the information contained in its explanations. As such, insofar as we care about decision-makers retaining meaningful control over all AI-supported decisions, it does not seem warranted for them to, at any point, stop monitoring the system's explanations.

However, it is also unhelpful to uncritically assume that knowing comes from merely looking (Ananny & Crawford, 2018). For a decision-maker to come to a well-reasoned assessment about whether to accept an AI-DST's recommendation, they must critically reflect about what its explanations imply in relation to broader social, political, and technical contexts surrounding the system's design and implementation. Decision-makers, then, should not merely "look into" their AI-DST, but also look across them – "[to see] them as sociotechnical systems that do not contain complexity but enact complexity by connecting to and intertwining with assemblages of humans and non-humans" (Ananny & Crawford, 2018, p. 982). Therefore, for a decision-maker to spot biases in their AI-DSTs, they must reflexively connect information contained in its explanations – about the system's hidden and immaterial layers of design – to broader social and economic structures, and in turn, to the injustice, oppression, and exploitation generated within these structures (Hollanek, 2020). It would follow, then, that critical reflection should also be an important aspect of how decision-makers interact with their AI-DSTs.

The particular mode described above – of interacting with an AI-DST through proactive monitoring and critical reflection – is what we are calling here a "deliberative attitude". The need for a deliberative attitude emerges in response to the opacity that characterizes AI-supported decision-making, and as such, seem to be more demanding and rigorous – in terms of both frequency and cognitive complexity – than ordinary calls to conduct routine "audits" and periodic "maintenance checks" of workplace tools (Clark et al., 1988; Grint & Woolgar, 2013). Indeed, different types of AI systems require different kinds of scrutiny depending on the underlying model architecture (cf. Arrieta et al., 2020), and there are a variety of informational cues – model parameters, intended use, evaluation data, training data, performance metrics, etc. – that a proactive 'human in the loop' would need to carefully scrutinize in their interactions with an AI-DST (cf. Mitchell et al., 2019). For this reason, we suggest, to ensure that the explanations from AI-DSTs meaningfully serve the pursuit of a broad range of normatively desirable goals beyond trust, it is crucial for decision-makers to adopt and maintain a deliberative attitude towards their AI-DSTs.

4.2 Trust and the Unquestioning Attitude

In Sect. 2.2 above, we highlighted a widely-accepted distinction between trust and mere reliance, and outlined some of the difficulties with conceptualizing what it means to trust – in a descriptively rich sense of the term – artifacts. In light of this distinction and its accompanying difficulties, we need to conceptualize trust in a way that at once preserves the trust-reliance distinction, while also allowing for trust in artifacts. A recent – and in our view, rather convincing – proposal by Nguyen (2020) might help us do so.

Broadly, Nguyen accepts Baier’s distinction between trust and reliance, and her attitudinal heuristic for separating the two, but rejects the view that trust can only be directed towards agents. To build his case, he invites us to consider what we would feel like if we were using a sturdy-looking rope to scale up a wall, and halfway through our journey, the rope suddenly breaks. Suppose, prior to this climb, we had carefully inspected the rope and found it to be of high quality, and that we had no reason to doubt that it was fit for purpose prior to its breaking. In this case, to say that we would feel merely disappointed when the rope breaks seems like an understatement. Indeed, the sharp shock and distress seems closer to some form of betrayal. Nguyen thinks this is because when we trust the rope, we are not bothering to consider whether it would hold steady (or fail) – we simply use it for our purposes while focusing our cognitive resources on other matters (p. 9). Trust, in this sense, is an “unquestioning attitude” – it emerges when we put questions about the rope’s reliability out of mind. Conversely, if we were to distrust the rope, we would climb carefully and re-examine it closely at every step. In this case, we might still be (forced to) rely on the rope, but “[this] reliance is tentative, and demands constant reassurance” (p. 9). If the rope were to give way now, we might feel disappointed, but not betrayed. In this way, Nguyen thinks we can preserve Baier’s distinction between trust and mere reliance even when talking about objects that are clearly non-agential.⁹

Although Nguyen’s account is not explicitly focused on AI systems, his view can help us understand what it would mean for a decision-maker to have full-blooded trust in their AI-DST. When a decision-maker adopts an unquestioning attitude towards their AI-DST – when they are not disposed to actively interrogate its reliability, and do not suspect that the system will unpredictably change its behavior – they trust the system.¹⁰ This does not entail that they will never question the system at all,

⁹ It is important to point out that Nguyen’s account, as well as other similar views (e.g. Ferrario et al.’s (2022) discussion of simple trust and the ‘anti-monitoring’ stance), do not exhaustively characterize what it means to trust technological artifacts, but simply describe – as fully as possible – the attitudes and dispositions that a trustor has towards their trustee. This is necessary, but perhaps insufficient, for fully describing trust in AI. A more complete conceptualization might need to also describe other beliefs, normative expectations, or assumptions that the typical trustor and trustee must have in relation to one another. Ferrario et al.’s (2022) “incremental model of trust” is one such careful and detailed attempt to do so. However, since the arguments in this paper are primarily centered on the *attitudinal* tensions between explainability and trust in AI, it is most important for us to clearly and completely characterize the attitudes that decision-makers have towards their AI-DSTs. For this, in our view, Nguyen’s “unquestioning attitude” view is especially instructive.

¹⁰ Importantly, for the purposes of our paper, Nguyen’s view allows us to remain neutral about the *object(s)* of trust. One may adopt an unquestioning attitude towards the AI-DST itself, or towards the sociotechni-

but rather, their trust simply entails the *disposition* to not question it. This disposition would be overturned and trust would be lost if the system were to suddenly malfunction in an obvious way, or if new evidence about the system's unreliability came to light. But as long as the decision-maker is generally disposed to not question their AI-DST, they may be said to trust their system in a descriptively rich sense of the term.

When this happens, the AI-DST is, as Nguyen would put it, *integrated* into the decision-maker's agency: the decision-maker allows the system to be intimately connected to their decision-making process, and give its recommendations a "direct pipeline into [their] cognitive and practical functioning" (p. 28). For decision-makers working with AI-DSTs, this integration is not only beneficial in terms of cognitive efficiency, but in some cases, a requirement of deploying such decision-making systems in environments where manual intervention is costly, challenging, or itself a source of risk. Consider, for example, the use of semi-autonomous rovers in space exploration missions. Here, operators are expected to allow their rovers' navigation systems to make thousands of calculations per second based on terrain conditions, features of explorative interest, weather conditions, etc. – only intervening to correct errors and/or take over navigation under particularly tricky conditions. Since, of course, the operator cannot explore Martian terrain themselves, and communication between the rover and operator can take minutes or even hours, they must depend on the rover to carry out the minutiae of mission objectives: trusting the rover's abilities to make split-second navigation decisions, while believing that the rover's onboard systems will correctly stop navigation and prompt for human intervention when necessary (Mandrake et al., 2022; Slingerland et al., 2022). It is this dynamic interrelation of agency and control, where the rover's decision-making capabilities and the operator's exploration goals are "knitted together into a functional whole", that Nguyen would call *agential integration* (p. 27).¹¹ Calls for trust in AI across a variety of domains – including defense and intelligence (U.S. Department of Defense, 2020; DNI, 2020), and organizational management (Gillespie et al., 2020; Glikson & Woolley, 2020) – often appeal to similar intuitions about decision-makers fully "accepting" and integrating AI-DSTs into their decision-making process, gesturing at the basic form of trust in artifacts that Nguyen describes, albeit often in different terms. Of course, the process through which this sort of trust might be obtained is often lengthy and highly involved – but once obtained, entails that decision-makers would come to adopt an unquestioning disposition towards their AI-DSTs.

cal system surrounding the AI-DST (including, for instance, the people and organizations that work to develop, deploy and manage the system) – but the disposition to put questions about the trustee's reliability out of mind remains, at a descriptive level, the same.

¹¹ On Nguyen's view, therefore, agential integration between the trustor and trustee is what makes 'betrayal' (rather than, say, anger or shock) the appropriate affective response when the trustee fails to do what they are trusted to do. It falls outside the scope of this paper to fully explain and evaluate the particulars of Nguyen's argument for this position, but we encourage interested readers to engage with his view in full, especially in Sect. 5 'The Integrative Stance' and 6 'Gullibility and Agential Outsourcing' (p. 24 onwards).

4.3 Attitudinal Tensions in the Joint Pursuit of Explainable and Trusted AI

As we have seen, for the explanations of an AI-DST to serve their intended normative ends, a decision-maker must adopt a deliberative attitude. And, for a decision-maker to trust their AI-DST in a descriptively rich sense of the term, they must adopt an unquestioning attitude. Although we have previously hinted at deep tensions between these two attitudes, we now make these tensions explicit. To do so, let us consider some simple – albeit extreme – scenarios.

Consider a decision-maker who, at some time t_I has a high level of trust in their AI-DST. This trust was established prior to t_I and could have been done so in a variety of ways: perhaps, as is often the case, they were directly and closely in the development process of this AI-DST and over time, they have come to believe that the system is especially well-tailored to augmenting their decision-making process.¹² Or, simply, perhaps their previous interactions with this AI-DST have been overwhelmingly positive. For whatever reason, from t_I onwards, the decision-maker adopts and maintains an unquestioning attitude towards their AI-DST. This entails that they put questions about the AI-DSTs reliability out of mind and are not disposed to actively monitor and scrutinize its output or explanations before accepting its recommendations. After t_I , they do not actively suspect that the system's recommendations could be biased or otherwise erroneous. However, active monitoring, scrutiny, and critical reflection are all crucial aspects of the deliberative attitude that is necessary for explainable AI to be morally useful. As we have argued above, for the system's explanations to meaningfully serve towards their intended moral goals, the decision-maker must actively monitor the content of these explanations, remain suspicious about the possibility of biases and other errors creeping in at any time, and maintain a careful sensitivity to the social, technical, and political context surrounding the AI system. Their unquestioning attitude makes such scrutiny unfeasible. The decision-maker's trust in the AI system disposes them to gloss over the complex and nuanced information contained in the content of its explanations, or perhaps even ignore these explanations entirely. As such, after t_I , the normative value of the system's explanations for making AI-supported decisions fair, accountable and legitimate appears greatly diminished.¹³

¹² For an AI-DST to be deployed in consequential decision-making processes, developmental milestones that have to be met often involve the negotiation of expertise between domain- and developer-knowledge. As recent ethnographic work in organizational studies indicates, these processes can be long and drawn-out, including both significant model compromise and extension (Kim & Mehrizi, 2022; Kim et al., 2022). It is, in our view, entirely feasible for individuals like Zoe to be involved in the development process through the translation of her clinical experience and expertise into model parameters. Resultantly, if Zoe believes that the resulting AI-DST is well-tailored to her needs and concerns, she is more likely to retain an unquestioning attitude in her interactions with this AI-DST that she has had significant involvement in training and deploying.

¹³ This is not to say, of course, that explanations are morally irrelevant at this point. For example, in case of harmful system malfunctions, explainability can be a valuable tool for retrospectively tracking how these malfunctions occurred, and how future similar malfunctions might be avoided. As such, even in cases where decision-makers no longer effectively scrutinizing their AI-DST's explanations (and in turn, failing to meaningfully perform their roles as 'humans-in-the-loop'), these explanations can still be morally useful in other ways. Our thanks to an anonymous reviewer for pointing out the need for this important clarification.

It is important to note that our argument is agnostic about *how* decision-makers come to trust their AI-DST (i.e., what happens before t_I). For this reason, we do not necessarily contradict the various empirical findings we discussed in Sect. 3 that show how explanations can serve to build trust in AI systems. Suppose instead, at some t_0 , the decision-maker is distrustful of their AI-DST, but upon carefully scrutinizing its explanations, they find its internal logic to be sufficiently robust, which then serves as a warrant for their trust. Over time, through continued interactions and scrutiny, they reach a state of high trust at t_I . That is, over time they no longer find it necessary to maintain continuous scrutiny of the system's explanations: they take their previous successful interactions with their AI-DST's output and explanations as inductive support for the system's continued trustworthiness. At t_I , then, they adopt an unquestioning attitude towards their AI-DST and start to spend less of their finite cognitive resources to question its reliability – perhaps choosing to be content with the mere existence of future explanations as a signal to reinforce their earlier decision to trust their AI-DST. At this point, once trust is stably secured, the normative value of the system's explanations once again diminishes: these explanations once again cease to play a meaningful role in making AI-supported decisions fair, accountable and legitimate. For this reason, even when explanations provide the basis for trust in AI-DSTs, the tensions between the unquestioning and deliberative attitudes persist.

Finally, let us return to Zoe from our earlier vignette. We know that she started off at a low-trust state at t_0 . She carefully scrutinized the system's output and explanations and learned that in some cases – i.e., when the AI-DST was prescribing medicines for a certain set of diagnoses – the system tended to be correct, and outside these cases, the system tended to get things wrong. Her scrutiny, therefore, did not provide her with a warrant for unqualified trust in their AI-DST. Rather, over time, as she gained a better understanding of the system's successes and failures, her trust came to be finely calibrated. In certain cases, where Zoe expected the AI-DST to perform well, she unquestioningly accepted its output, and in all other cases, she carefully deliberated about its output and explanations. By switching between unquestioning and deliberative attitudes depending on the situation, it might seem like Zoe is able to avoid the attitudinal tensions discussed above.

Not quite, we think. Trust in an AI-DST, even when finely calibrated, can still be in tension with the normative value of explanations. For one, AI-supported decision-making processes are typically dynamic: AI-DSTs often train and update their parameters continuously with new data, and the contextual factors surrounding a decision can also change in significant ways (Gama et al., 2014; Tsymbal, 2004). This dynamism means that, in general, AI-DSTs are unstable objects of trust – prior evidence about their reliability within certain domains may not be fully indicative of their future performance. New virus mutations, for example, could make Zoe's AI-DST less effective in its antiviral prescriptions, or a new update to her system might introduce a bias that did not exist previously. In such cases, Zoe's previously calibrated trust may no longer be warranted, and she would have to once again monitor and critically reflect about the system's output and explanations to recalibrate her trust. The more dynamic the decision-making context in question, the more often one would have to perform such recalibrations: but continuous recalibration requires the

continuous adoption of a deliberative attitude, which itself forecloses full-blooded trust.

Moreover, even when trust is finely calibrated, there remain difficult questions about holding decision-makers responsible for decisions that they have accepted unquestioningly. Suppose Zoe's AI-DST outputs a prescription for a certain medical diagnosis, and Zoe, based on her previous scrutiny, trusts that the system performs especially well when faced with this particular diagnosis. Unfortunately, there happens to be a rare failure: the decision is erroneous, possibly in a deeply harmful way. Oblivious to this fact, Zoe accepts the system's decision unquestioningly, without scrutinizing its explanations – and in turn, fails to prevent the ensuing harms. Now, one possible response to this case might be to suggest that Zoe's conduct was, all things considered, appropriate. Her trust was finely calibrated based on her careful previous scrutiny of the system and, despite this rare failure, her calibrated trust would allow her to efficiently produce correct decisions most of the time. But, we suspect, such a response would be deeply dissatisfying to those who would want to hold Zoe responsible for this particular erroneous decision's harms. The AI-DST's explanations should have provided her with enough information so that she could intervene in time to prevent its harmful decisions. Her decision to stop scrutinizing these explanations, and in turn, her failure to stop its harmful decision, seems to be reasonable grounds for holding her – as the so-called human-in-the-loop – accountable for the ensuing harms. As such, for those who argue for XAI on the basis that explanations are necessary for holding decision-makers responsible for harmful decisions, full-blooded trust – calibrated or otherwise – would seem deeply inappropriate.

With these illustrations, we are trying to show that it is profoundly difficult for any decision-maker to both trust their AI-DST while also responding in an appropriate way – through proactive monitoring and critical reflection – to its explanations. Unquestioning and deliberative attitudes are incompatible attitudinal states for any decision-maker to occupy simultaneously. For an explanation to be normatively useful, it must be carefully scrutinized, evaluated, and acted on. Full-blooded trust, which entails the adoption of an unquestioning attitude towards one's AI-DST, makes such scrutiny difficult. For this reason, the joint pursuit of explainable and trusted AI runs into an intractable *implementation gap*: these two principles, when put into practice, call upon decision-makers to adopt and maintain two distinct attitudes towards their AI-DST that are profoundly difficult to adopt simultaneously, and the adoption of either attitude supports the pursuit of one principle at the expense of the other. By sticking to the process model – where the two principles are defined only in terms of their instrumental value for increasing technology adoption and acceptance – we would likely fail to uncover these attitudinal tensions between explainability and trust. However, by adopting a more comprehensive view of what these two principles entail, it becomes clear that their joint pursuit will encounter, and in turn, must overcome a critical implementation gap.

5 Implications and Recommendations

At this point, it is important to clarify the scope and implications of our intended conclusion. We do not mean to suggest that the joint pursuit of explainable and trusted AI is somehow necessarily self-defeating, or that we must choose one of these principles to abandon entirely. Our two intended provocations are rather more modest. First, we suggest that the assumed straightforward connection between explainability and trust – where we care about the former only insofar as it leads to the latter – is inadequate, for it fails to account for the rich and varied normative reasons for why explainability and trust in AI matter. And secondly, we suggest that since the joint pursuit of these two principles entails the simultaneous adoption of incompatible attitudinal states, this joint pursuit is likely to be hindered by a thorny implementation gap between principles and practice.

Note, however, that this gap arises out of our theoretical conception of (a) why we care about the explainability of AI, and (b) what it means to trust AI in a descriptively rich sense. Of course, it is wholly possible to conceptualize trust and explainability in a different way, to avoid these attitudinal tensions altogether. For instance, alternative justifications for explainability might focus primarily on how explanations can benefit decision *subjects* rather than decision-makers. Such justifications might foreground the oft-discussed ‘right to explanation’: where, at least on certain readings of the European Union’s General Data Protection Regulation (GDPR), decision-subjects have the right to obtain meaningful information about the logic, significance, and expected consequences of an AI-supported decision (Edwards & Veale, 2017; Goodman & Flaxman, 2017). Vredenburg (2019), for instance, suggests that such a right might be grounded in the requirements of fairness, and particularly in the requirement for decision subjects to be able to engage in informed self-advocacy. Similar proposals focus on the decision subject’s ability to contest a decision, seek recourse for unfair outcomes, and give meaningfully informed consent to institutions that employ AI-DSTs (Edwards & Veale, 2017; Kaminski, 2021; Kim & Routledge, 2021; Ustun et al., 2019). If the moral reasons for pursuing XAI solely focus on the decision-subject, we might no longer need decision-makers to adopt and maintain the deliberative attitude, and in turn, pursue full-blooded trust without running into the implementation gap discussed above. Such a move might seem especially attractive to those who think that the cognitive burdens of scrutinizing explanations and being an effective ‘human in the loop’ are too demanding. But in turn, we might have to give up on various existing, and widely-defended, moral reasons for pursuing explainability – e.g., to hold decision-makers accountable, and to help them spot biases and errors. Given growing worries about algorithmic bias and unaccountable algorithmic decision-making, this option seems rather unattractive.

Alternatively, perhaps we could seek to conceptualize ‘trust’ in such a way that forecloses the adoption of an unquestioning attitude as a reasonable way for decision-makers to relate to their AI-DSTs. One way to do so might be to insist on a strict, *normative* distinction between trust and mere reliance. Here, one may argue that trust is strictly interpersonal (i.e., between human agents), and that decision-makers may at most *rely* on their AI-DSTs. If only reliance is desired – and all that matters is the probability of the decision-maker accepting their AI-DST’s correct recommenda-

tions and rejecting its incorrect ones – there is no requirement for them to maintain an unquestioning attitude towards their AI-DST. In fact, maintaining a deliberative attitude might even be conducive to effective reliance. Decision-makers that proactively monitor and critically reflect about their AI-DST's output and explanations would be especially well-placed to make accurate decisions about when to accept or reject their system's recommendations. Moreover, existing scholarship in the fields of management, information systems, and human-computer interaction has explored how "appropriate reliance"¹⁴ relates to the acceptance and adoption of technological systems (Wu et al., 2011). As such, insofar as we are concerned about increasing the adoption of AI-DSTs by decision-makers, perhaps a focus on reliance is sufficient. Having said that, as we have argued above, trust and reliance differ not only normatively, but also descriptively. It seems reasonable to think that decision-makers could adopt similar attitudes and dispositions towards a trusted AI as they would towards a human they trust – and at least sometimes, this 'trust' (in a descriptively rich, but non-normative sense of the term) might be desirable. A pressing worry is that solely focusing on reliance would entail giving up on "agential integration" – the state of synergy, as Nguyen describes it, where the decision-maker allows their AI-DST to be intimately connected to their decision-making process. It seems reasonable to think that decision-makers should do more than just rely – cautiously and tentatively – on their AI-DSTs. Rather, we hope that through rich and synergistic user-AI collaborations, we might move towards decision-making processes that make the best use of both algorithmic data processing and human judgement. It is unclear how, if at all, mere reliance can help decision-makers to pursue these desirable synergies with their AI-DSTs.

In general, we recognize that the general strategy pursued so far – of sidestepping our identified implementation gap by simply tweaking our concepts of explainability and trust – might seem rather disappointing. Still, we thought it was important to discuss, albeit briefly, what this strategy entails. In part, we wish to highlight how our preceding analysis of the attitudinal tension hinges on the – reasonable, but still contested – conceptualizations of explainability and trust that we have adopted in this paper. We also remain open to the possibility that there might be ways to conceive of trust and explainability that accord with our various intuitions about these concepts, while also avoiding the attitudinal tensions discussed above. We wish to invite future theorists to develop such concepts, and hope that our brief discussion in this section so far might help towards their theorizing.

In the remainder of this section, allow us to explore another strategy to move beyond the implementation gap in the joint pursuit of these two principles – one with more straightforwardly practical implications. At the heart of this strategy is the recognition that this gap arises from the difficulties of asking any *single* decision-maker to simultaneously adopt both unquestioning and deliberative attitudes towards their AI-DST. But, in a more complex decision-making chain, it might be possible

¹⁴ As we mentioned earlier, a large amount of empirical literature on this topic holds 'trust' as conceptually equivalent to 'appropriate reliance'. Those who wish to, as we suggest, press a strict normative distinction between trust and reliance would need to mount an effective challenge against those who argue for 'trust as appropriate reliance'. Mark Ryan's (2020) paper is one notable attempt to do so, and his careful and detailed arguments may be instructive to those who wish to pursue this strategy.

for different groups to maintain each of these attitudes separately. In practice, decision-making chains can be arranged so that someone (or some group of people) is explicitly charged with monitoring and critically analyzing the AI-DST's output and explanations – i.e., by maintaining a deliberative attitude – so that others may take the fact that the system is being scrutinized as their basis for adopting an unquestioning attitude. Put differently: *some deliberate, so that others can trust.*

This general strategy may be implemented in various ways. For one, organizations may explicitly task the so-called “human-in-the-loop” – the person who directly interfaces with the AI-DST – to constantly scrutinize and deliberate about the system's recommendations and explanations. These intermediaries would, in turn, pass on only those recommendations that they deem appropriate to a decision-maker, who would make the final decision. In such an arrangement, it would clearly be undesirable for the human-in-the-loop to stop maintaining the deliberative attitude at any time. However, higher-level decision-makers may, on the basis that someone has already done the work to carefully select the system's appropriate recommendations, unquestioningly accept these selected recommendations. Similarly, this intermediating function might also be performed by a group of people – organizations may charge a so-called “red-team” with purposefully maintaining a deliberative (or perhaps even adversarial) stance towards their AI-DST and trust the system on the basis that this team's scrutiny and deliberation is adequate and reliable. In such a way, some decision-makers might be able to enjoy the cognitive benefits of agential integration with their AI-DST, while ensuring that the system's explanations are carefully monitored and analyzed simultaneously.

Certainly, such arrangements may not be feasible in all decision-making contexts. In clinical settings, to take one example, decision-support systems are meant to be used directly by doctors and other medical professionals, leaving little room for the meaningful involvement of any intermediary.¹⁵ Moreover, it might sometimes be prohibitively expensive and inefficient for organizations to set up internal teams exclusively to scrutinize their AI-DSTs without making any decisions themselves. In such situations, one alternate strategy – that not only represents an efficient process, but might also add additional legitimacy to the use and implementation of these tools overall – might be to outsource the responsibility of deliberation and scrutiny to *external* institutions.

In the automotive and aviation industry in the United States, for example, external, governmental bodies such as the National Highway Traffic Safety Administration or the Federal Aviation Administration respectively carry out research to identify current and future issues, set and harmonize safety standards, and enforce or audit internal processes. Further, these institutions, by merit of being external to and inde-

¹⁵ It is useful to note that, in some cases, AI-DSTs that impose significant liability burdens on individual decision-makers might represent a good target for regulatory interventions. This is especially the case in domains – such as clinical decision-making – that fall clearly under purview of a powerful and well-resourced regulator, who might be able to intervene in reviewing and validating AI-DSTs, and properly calibrating and demarcating the professional liability of decision-makers who use these systems. Our thanks to an anonymous reviewer for pointing out that Zoe's AI-DST for clinical prescriptions, introduced earlier in Sect. 4.3, would be a good regulatory target, and if so, it is possible that Zoe might not be held personally liable for harms ensuing from the AI-DSTs potential errors.

pendent of the companies they oversee, increase consumer trust in the products and services they use. These potential benefits have not gone unnoticed in policy and regulatory conversations about AI. Already, attempts to establish similar external auditing and standards-setting institutions for AI are being pursued by governments (Infocomm Media Development Authority, 2021; UK Information Commissioner's Office, 2019), corporations (Hao, 2021) and academics (Brown et al., 2021; Koshiyama et al., 2021). For the moment, the impact and efficacy of these attempts remain rather limited (Sloane et al., 2021), but with enough time and resources, it seems possible to obtain credible and well-resourced external institutions that can scrutinize and monitor AI-DSTs for organizations, thereby allowing decision-makers to fully trust their systems.

Yet, external institutions should not be seen as a panacea for overcoming the deep-seated tensions in the joint pursuit of explainable and trusted AI. Crucially, these institutions themselves need to continuously demonstrate their independence and trustworthiness. The regulatory capture of these entities, for instance, has the potential to not only decrease competitiveness and innovation amongst industry competitors, but also might increase the potential harms faced by consumers. These are *governance* problems, inherent to most configurations of state/market solutions. Moreover, there are also *practical* problems that constrain the efficacy of such institutions. For one, organizations can be unwilling to provide sufficient information – about their AI-DST's outputs and explanations, and the organizational and social context of its deployment – to third-party auditors, rendering their scrutiny ineffective (Sloane et al., 2021). Additionally, as we alluded to earlier, decision-making contexts are often dynamic, and AI-DSTs can change substantially over time. As a result, the state of an AI-DST that decision-makers decide to trust can be quite different from the state that was previously scrutinized and certified by an external institution. This dynamism calls for continuous monitoring and scrutiny, which can once again be prohibitively resource-intensive.

To take stock, in this section we have sketched a few plausible – but imperfect – strategies for overcoming our theorized implementation gap. The fact that no obvious strategy presents itself should not be surprising, since the problem itself, to our best knowledge, has been relatively underexplored so far. Even so, we hope that our brief discussion of the benefits and shortcomings of these different strategies can stimulate further research about what precisely is needed to facilitate the joint pursuit of explainable and trusted AI in decision-making contexts.

6 Conclusion

The joint pursuit of explainability and trust in AI-DSTs is widely held to be crucial for the promotion of ethical and socially beneficial AI. However, we have argued that since these two principles entail the adoption of attitudinal states that are incompatible with one another, there seems to be a critical implementation gap in bringing these principles jointly to practice. Dominant assumptions about how the two principles relate to one another – i.e., that explainability leads to trust, and trust leads to adoption – obfuscate this gap. However, if we take into account the rich and diverse

moral reasons that ground the call for explainable AI, we can see that a meaningful pursuit of the principle of explainability entails that decision-makers should adopt and maintain a deliberative attitude towards their AI-DSTs. Similarly, if we consider what it means to trust – in a way that goes beyond mere reliance on – an AI-DST, we can see that full-blooded trust entails that decision-makers should adopt and maintain an unquestioning attitude towards their AI-DSTs. We then showed that, if a decision-maker is expected to both trust their AI-DST *and* scrutinize its explanations, they would be expected to adopt and maintain incompatible attitudinal states. The challenge, we suggest, is to find ways to both preserve our normatively full-blooded conceptions of these two principles, while also avoiding making incompatible demands about the appropriate attitude that decision-makers should adopt towards their AI-DSTs.

Importantly, the attitudinal tensions that we have identified between these two principles emerge out of our specific theoretical conceptualizations of why explainability matters, and what it means to trust AI. To fully understand the nature of these attitudinal tensions and how exactly they play out in real-world decision-making contexts, we would need further empirical investigations into the psychology of trust and deliberation in relation to AI-DSTs. We invite future researchers to pursue such empirical research, and hope that the concepts we have developed here may aid in their efforts. Moreover, another germane line of future research might be to investigate questions of internal and external governance structures, to identify the conditions needed to support the effective scrutiny and analysis of explanations from AI-DSTs, so that decision-makers are freed-up, so to speak, to fully trust their systems.

As such, our discussion of the attitudinal tensions between explainable and trusted AI ultimately means to support, rather than undermine, the joint pursuit of these two critical principles. Our hope is that, through concerted future empirical and theoretical research, we might uncover effective strategies to mitigate these tensions, and in turn, meaningfully advance the implementation of ethical and socially beneficial AI.

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Declarations

Conflict of Interest The authors have no competing interests to declare that are relevant to the content of this article.

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